Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/21/2018
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ferpello@me.berkeley.edu	**Fax:**	FY 510 642-6163
PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
Organization Name:	University of California, Berkeley		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	6105A Etcheverry Hall		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	05/21/2013	**End Date:**	10/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2020 per PI and S. Olson/GRC (Ed., 3/23/16) NOTE: Extended to 10/31/2015 per NSSC information (Ed., 3/24/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AL10A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres. Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. Although the most effective fire safety strategy is to prevent ignition, if ignition does occur, it is critical to rapidly extinguish the fire. Therefore, an experimental investigation into ease of ignition and fire suppression characteristics of materials under SEA is proposed. Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion Integrated Rack on the International Space Station (ISS). It consists of a small-scale cylindrical flow duct that has fuel samples attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA. Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, suppression, and spread characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
Task Progress:	During this period, the flame spread and mass burning of cylindrical PMMA samples were studied under varying opposed flow, pressure, O2 concentration, and external radiation. Some of the results were compared with microgravity data obtained in the BASS II experiments. One of the objectives of the work determine the transition between opposed flame spread and mass burning (or fuel regression). Results show that in normal gravity, as the opposed flow increases to 50~100 cm/s, the flame can no longer spread over the fuel surface, but becomes a mass burning process, staying in the recirculation zone downstream of the cylinder top surface. Thus, the observed “flame-spread” is actually a fuel-regression like a pool fire flame or a candle flame. The fuel regression only occurs for the thick rods, but not for the thin rods which do not have a downstream recirculation zone. Such transition can also be indicated by a critical leading-edge regression angle of ˜45o. Increasing the opposed flow velocity, the shape of cone becomes flatter until flame blow-off occurs. In the present experiments, before blow-off, a flat blue flame is formed floating above the rod top s urface, like a classical counterflow diffusion flame. Such blue flame is stable and is a result of vortex shedding. As the ambient pressure or O2 concentration decreases, the flame spread first transitions to the fuel regression regime, and then, extinction occurs (two-step extinction process). On the other hand, external radiation in the perpendicular to the flame-spread direction may prevent the transition to fuel regression if the radiant flux is elevated enough. In microgravity, the fuel regression angle is found to be much smaller (<5o) in the low opposed flow (<10 cm/s) and the cone-shape fuel may not be formed after the flame spread, which is different from normal gravity. This work not only provides a better understanding of flame-spread and blow-off phenomena under various environmental conditions, but also clarifies the differences between the flame-spread and fuel-regression on solid fuels.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Articles in Peer-reviewed Journals	Link S, Huang X, Fernandez-Pello C, Olson S, Ferkul P. "The effect of gravity on flame spread over PMMA cylinders." Scientific Reports. 2018 Jan 9;8(1):120. https://doi.org/10.1038/s41598-017-18398-4 ; PubMed PMID: 29317681; PubMed Central PMCID: PMC5760683 , Jan-2018
Papers from Meeting Proceedings	Huang X, Link KS, Rodriguez A, Olson S, Ferkul P, Fernandez-Pello C. "Transition from Opposed Flame Spread to Fuel Regression: Effect of Flow, Atmosphere, and Microgravity." 36th International Symposium on Combustion, COEX, Seoul, Korea, July 31 - August 5, 2016. 36th International Symposium on Combustion, COEX, Seoul, Korea, July 31 - August 5, 2016. , Aug-2016
Papers from Meeting Proceedings	Hernandez N, Reszka P, Fuentes A, Fernández-Pello C. "Piloted Ignition Delay Times of Optically Thin PMMA Cylinders." 36th International Symposium on Combustion, COEX, Seoul, Korea, July 31 - August 5, 2016. 36th International Symposium on Combustion, COEX, Seoul, Korea, July 31 - August 5, 2016. , Aug-2016